

DTOF RGB-D CAMERA · S SERIES

COMPACT MRDVS S10 RGB-D CAMERA

A 190-gram industrial RGB-D camera that returns a valid depth value on every pixel – black rubber, polished steel, or a blank white wall – from 0.3 to 8 metres, indoors or in full sun.



■ WHY THIS CAMERA

- **Direct time-of-flight depth** – every pixel returns a valid range regardless of surface texture, colour or contrast
- **0.3–8 m at 90% reflectivity**, 0.3–3 m at 5% reflectivity, with ≤ 3 cm accuracy held across the full working range
- **120° × 80° depth field of view** – the tallest vertical FOV in the S Series, for low-obstacle coverage from a chassis mount
- **100 kLUX ambient light immunity** – stable through direct sunlight, floor-to-ceiling glass, and dark warehouse aisles
- **Depth and RGB aligned in-camera** – outputs depth, IR amplitude, RGB, point cloud and processed obstacle-avoidance results
- **≤ 4 W average draw** at 80 × 37 × 25 mm and 190 g – sized for an AMR chassis, not a test bench
- **100M Ethernet with PTP time sync** and a C / C++ / ROS 1 / ROS 2 SDK for Windows, Linux and Arm Linux
- **IP54, -20 °C to 60 °C**, Class 1 laser per IEC 60825-1 – eye-safe in normal use

0.3–8 m

WORKING RANGE

120° × 80°

DEPTH FOV

≤ 3 cm

ACCURACY

190 g

WEIGHT

■ APPLICATION SCENARIOS



AMR & AGV CHASSIS

Forklift and humanoid chassis
obstacle avoidance



LAST-MILE DELIVERY

Sidewalk and curb perception at low
speed



PARCEL DIMENSIONING

Four-camera DWS volumetric
measurement



PASSENGER FLOW

Turnstile tailgating and people
counting

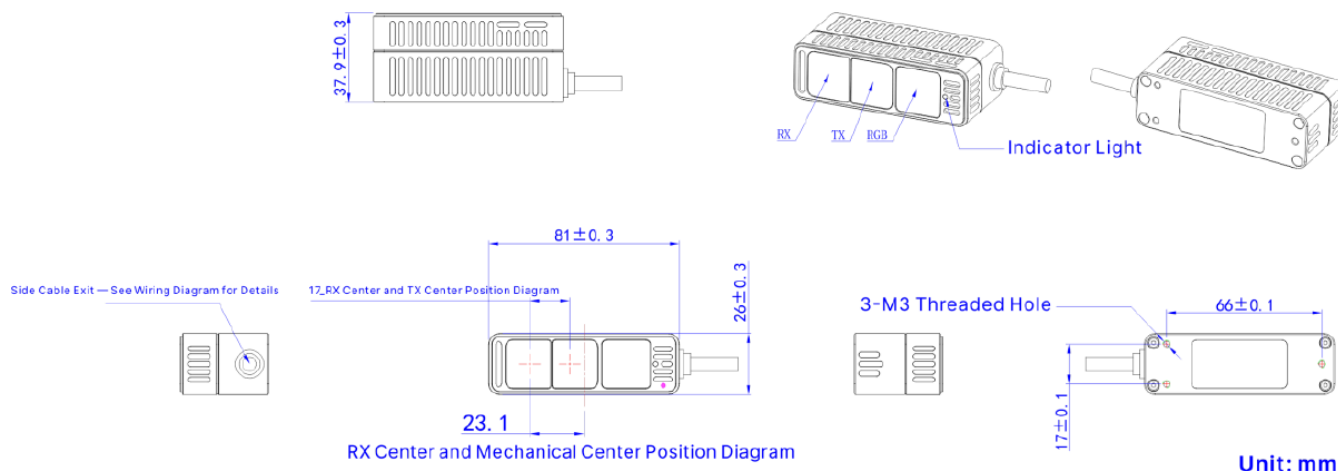
MRDVS S10 – SPECIFICATIONS



PARAMETER	TECHNICAL DATA
Sensing technology	dToF (direct time-of-flight) SPAD array
Laser wavelength	940 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Output format	Depth / RGB / IR amplitude map / point cloud
Depth resolution & frame rate ^①	240 × 160 @ max. 20 fps (typ. 15 fps)
ToF FOV (H × V)	120° × 80°
RGB resolution & frame rate ^①	1632 × 1224 @ max. 20 fps (typ. 15 fps)
RGB FOV (H × V)	120° × 80°
Range	0.3–8 m (90% reflectivity) 0.3–3 m (5% reflectivity)
Accuracy ^②	≤3 cm
Ambient light resistance	100 kLUX
IMU	—
Interface	100M Ethernet

PARAMETER	TECHNICAL DATA
MIPI CSI-2 available	No
Time synchronisation	PTP
Power supply	12–28 V DC
Average power consumption	≤4 W
Dimensions (L × W × H)	80 × 37 × 25 mm
Weight	190 g
IP rating	IP54
Operating temperature	–20 °C to 60 °C
Storage temperature	–40 °C to 85 °C
Humidity	Max. 90% RH, operating and storage
SDK support	C / C++ / ROS 1 / ROS 2
OS compatibility	Windows 7/8/10/11, Linux, Arm Linux
GUI software	LxCameraViewer

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. CE, Class 1 laser safety (IEC 60825-1:2014), RoHS (2011/65/EU Annex II and amendment (EU) 2015/863), REACH (EC No 1907/2006). EMC per EN IEC 61000-6-2:2019 and EN IEC 61000-6-4:2019.

Notes. ① Typical (typ.) frame rate is the recommended operating condition for optimal thermal and power performance; maximum (max.) frame rate is technically achievable but increases power draw and temperature rise and is not recommended for general use. Contact MorpheusTEK for application-specific evaluation. ② Accuracy may degrade with distance and depends on object reflectivity and surface roughness.

Installation. Mount the camera centred in its opening with no housing edge, rib or bracket intruding into the nominal field of view; FOV truncation causes fixed blind areas and false close-range readings. Keep the sensor at least 10 cm from highly reflective surfaces. Apply moderate, even mounting torque. Provide ventilation in enclosed spaces. Clean the optical window on a regular schedule in dusty or humid environments.

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